



EX1.

a.

the strategy of A is $\{L, R\}$

strategy of B is $\{(l, l), (l, r), (r, l), (r, r)\}$

Therefore, strategic form of this game shown as follow:

A \ B	(l, l)	(l, r)	(r, l)	(r, r)
L	u_1	u_1	u_2	u_2
R	u_3	u_1	u_3	u_1

b.

saddle points are $\{L, (l, r)\}, \{R, (l, r)\}$

c.

the value of this game is u_1

EX2.

the binary representation of the initial board is

k	number	1	2	4	8
1	1	1			
2	3	1	1		
3	7	1	1	1	
4	15	1	1	1	1

the board is unbalance, therefore the first player is winner if ze playing optimally

at first round, first player pick $8 + 2 = 4$ matchsticks from pile 4 to make it balance, the board become

k	number	1	2	4
1	1	1		
2	3	1	1	
3	7	1	1	1
4	5	1	0	1

the second take all matchstickers from first pile, the board become

k	number	1	2	4
2	3	1	1	
3	7	1	1	1
4	5	1	0	1

at second round, first player pick 1 from pile 4, the board become

k	number	1	2	4
2	3	1	1	
3	7	1	1	1
4	4	0	0	1

second player pick pile 2, board become

k	number	1	2	4
3	7	1	1	1
4	4	0	0	1

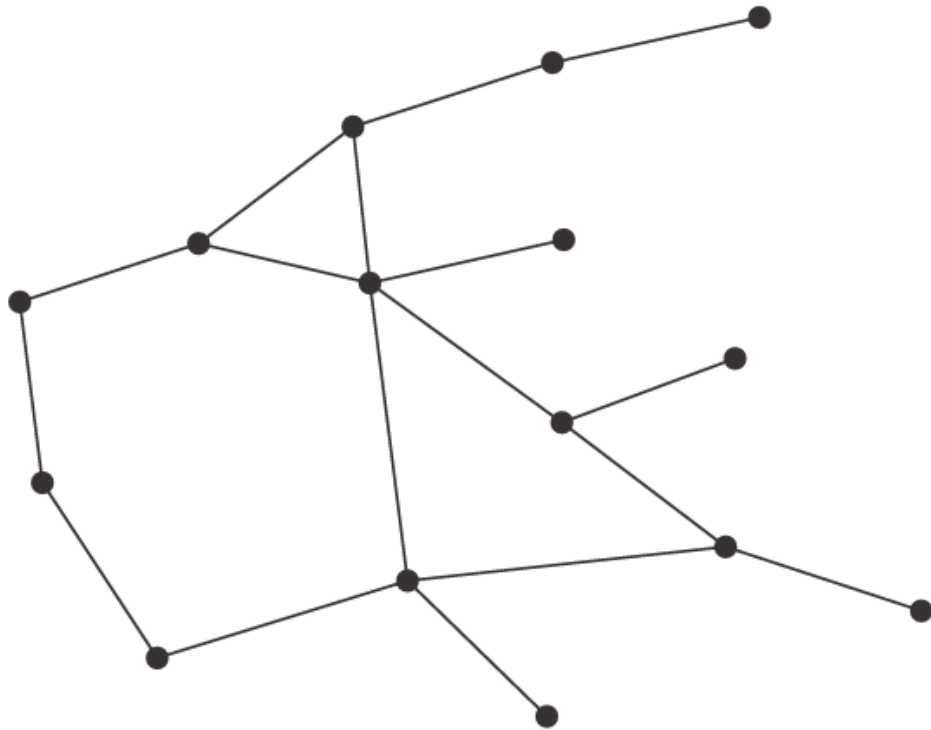
at third round, player 1 pick 3 from pike 3, board become

k	number	1	2	4
3	4	0	0	1
4	4	0	0	1

second player pick arbitrary, and then the first pick remaining, the first player win.

EX3.

15. Two players alternately remove nodes from a connected graph $\mathcal{G}$. Except in the case of the first move, a player may remove a node only if it is joined by an edge to the node removed by the previous player. The player left with no legitimate vertex to remove loses. Explain why the second player has a winning strategy if there exists a set E of edges with no endpoint in common such that each node is the endpoint of an edge in the set E . Show that no such set E exists for the graph of Figure 2.22. Find a winning strategy for the first player.



- definition in convenience

We called the set which satisfying the condition is a **perfect matching**

- the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is vertices set and $\mathcal{E}$ is edges set

proof of the winning condition

first, we claim the second player's strategy is:

- if an perfect set $E \subseteq \mathcal{E}$ exist in the first player's turn, assume the first player take one node $p \in \mathcal{V}$, by definition there exist only one edge in E (calling e) with endpoint P .
assume the edge e connect p and q , then the second player's strategy is take q .
- for the case E does not exist in the first player's turn, second player randomly pick one legal node from $\mathcal{V}$

then we can prove by induction that the second player win by strategy above if the perfect set exists.

- first, consider the case that E have only one edge.
at this case, $\mathcal{G}$ only have 2 nodes and one edge, the first player take one, the second take the remaining, and nothing left for the first, therefore ze lose.
- assume the claim is true for $|E| = n = \frac{\|\mathcal{V}\|}{2}$, then for the case $|E| = n + 1$:
following the strategy above, after the second player's action, all vertices of e are removed, and we can find that nodes in $E - \{e\}$ will cover $\mathcal{V} - \{p, q\}$ again.

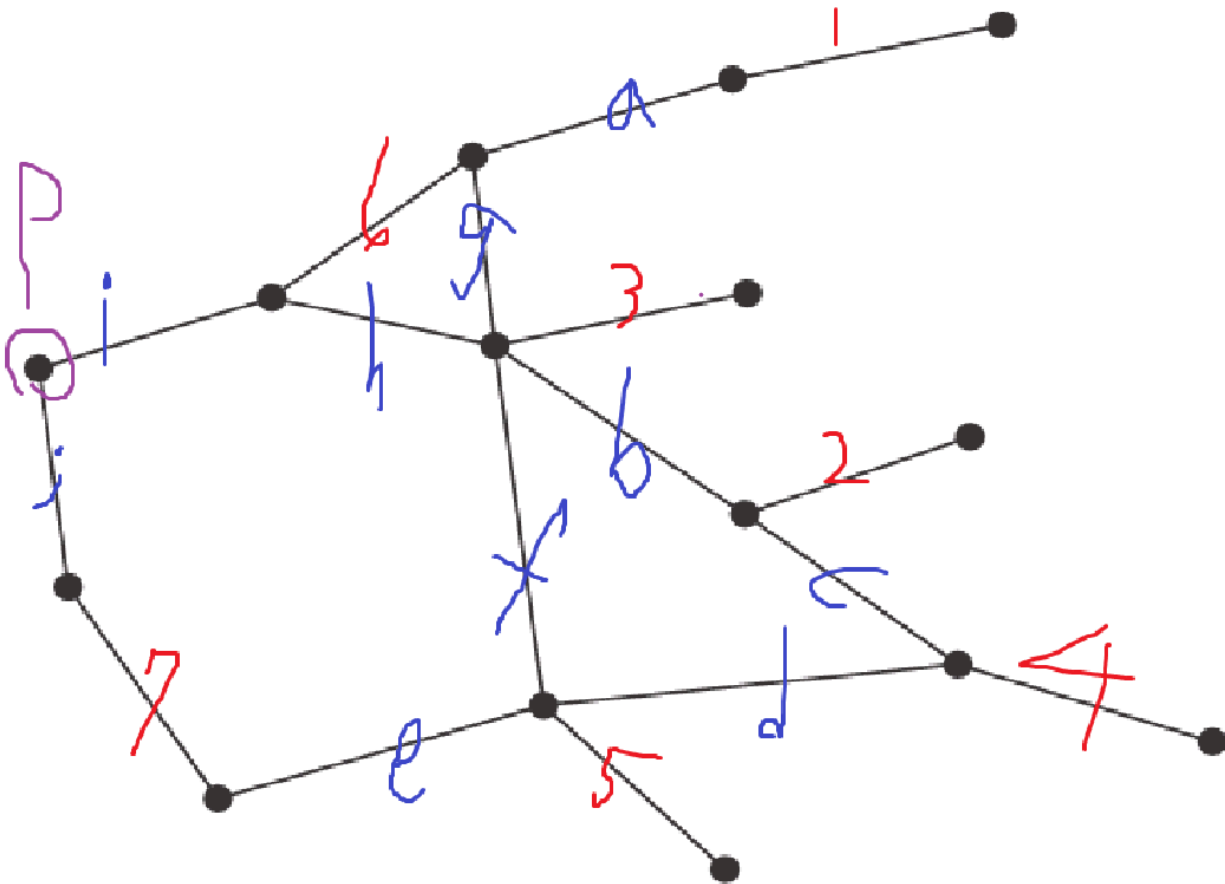
In other words, after the second player's turn, the graph $\mathcal{G}'$ will have a perfect set $E' = E - \{e\}$ with size $|E'| = |E| - 1 = n$.

by induction hypothesis, the second player can win this game.

proof E does not exist

prove by contradiction that E does not exist.

Let's indexing the edge as follow



first, we observe all nodes which only have one connected edge. because E need to cover all vertices, therefore E must include all corresponding edges. In other words, $\{1, 2, 3, 4, 5\} \subseteq E$.

Because edges in E cannot have common endpoints, we can find that $\{a, b, c, d, e, f, g\} \cap E = \emptyset$

with the same argument of the first step, we can find that $\{6, 7\} \subseteq E$, and by the same argument of the second step, $\{i, j\} \cap E = \emptyset$

therefore, we can find that $E = \{1, 2, 3, 4, 5, 6, 7\}$ if exist. however, the set does not cover P (the purple one), contradict with the definition of E , and therefore E does not exist.

winning strategy of the first player

The first player pick up P (the purple one) in the graph, and then

- it's the second player's round

- $\{1, 2, 3, 4, 5, 6, 7\}$ become a perfect set of the graph after removing P

then the first player apply the strategy of previous subproblem (but the character is swapped), then ze can win the game as mentioned above.

EX4.

19. Let (s, t) and (s', t') be two different saddle points for a strictly competitive game. Prove that (s, t') and (s', t) are also saddle points.

Let $v(a, b)$ denote the value of the game in strategy (a, b)

from the definition of saddle points, we have

- $v(s, t) = v(s', t') =$ the value of this game
- $v(s, t) \preceq v(s, \hat{t}) \forall \hat{t} \in \text{strategy of player } B \text{ --- (1)}$
- $v(s, t) \succeq v(\hat{s}, t) \forall \hat{s} \in \text{strategy of player } A \text{ --- (2)}$
- $v(s', t') \preceq v(s', \hat{t}) \forall \hat{t} \in \text{strategy of player } B \text{ --- (3)}$
- $v(s', t') \succeq v(\hat{s}, t') \forall \hat{s} \in \text{strategy of player } A \text{ --- (4)}$

Therefore, $\forall \hat{t} \in \text{strategies of player } B$, we have

- $v(s, t') \stackrel{(3)}{\preceq} v(s', t') = v(s, t) \stackrel{(1)}{\preceq} v(s, \hat{t})$
- $v(s', t) \stackrel{2}{\preceq} v(s, t) \stackrel{0}{=} v(s', t') \stackrel{(4)}{\preceq} v(s', \hat{t})$

By the similar step, for any $\hat{s} \in \text{strategies of } A$, we have

- $v(s, t') \succeq v(s, t) = v(s', t') \succeq v(\hat{s}, t')$
- $v(s', t) \succeq v(s', t') = v(s, t) \succeq v(\hat{s}, t)$

Therefore, by definition, (s, t') and (s', t) are saddle points.

EX5.

26. The rhyming triplets, Boris, Horace, and Maurice, are the membership committee of the very exclusive Dead Poets Society. The final item on their agenda one morning is a proposal that Alice should be admitted as a new member. No mention is made of another possible candidate called Bob, so an amendment to the final item is proposed. The amendment says that Alice's name should be replaced by Bob's. The rules for voting in committees call for amendments to be voted on in the reverse order to which they are proposed. The committee therefore begins by voting on whether Bob should replace Alice. If Alice wins, they then vote on whether Alice or Nobody should be made a new member. If Bob wins, they then vote on whether Bob or Nobody should be made a new member. Figure 2.24(a) is a diagrammatic representation of the order in which the voting takes place. Figure 2.24(b) shows how the three committee members rank the three possible outcomes.

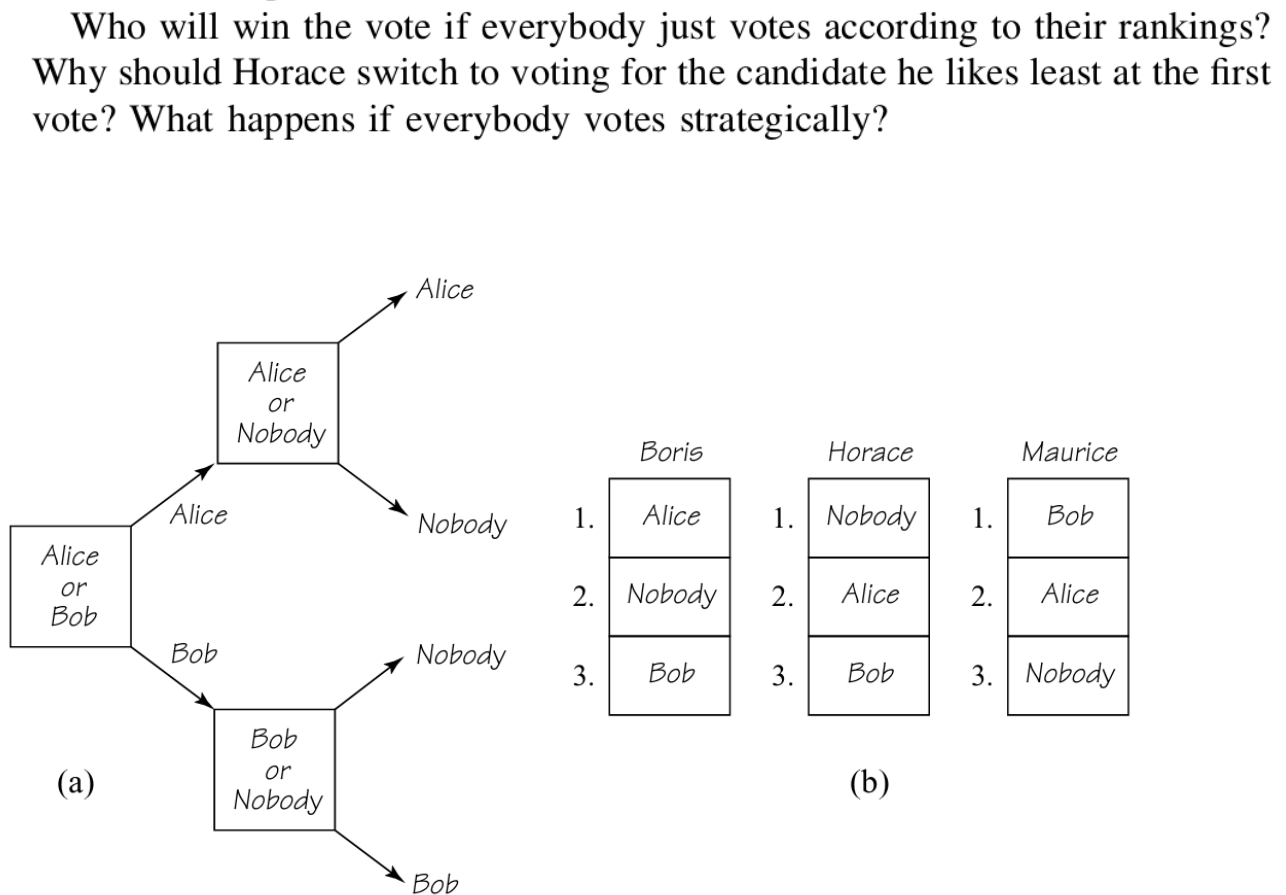


Figure 2.24 Strategic voting.

if they vote only by ranking

result of the vte shown as below

round	Alice	Bob	Nobody
1	2	1	$\emptyset$
2	2	$\emptyset$	1

Alice will win

strategy of Horace

If Horace vote Bob at first round, vote Nobody at second round, and others still vote by ranking, then the result become

round	Alice	Bob	Nobody
1	1	2	$\emptyset$
2	$\emptyset$	1	2

Nobody will win, which is Horace hope the most.

if the vote in strategy

at now, we know the result of the second round of each case:

- if Alice vs Nobody, Alice win
- if Bob vs Nobody, Nobody win

and we know the preference of each voter, therefore

- Boris hope Alice win at first round
- Horace hope Bob in at first vote
- Maurice know Bob cannot be the winner, so he will hope Alice win

Therefore, at first round:

- Boris and Maurice vote for Alice

- Horace vote for Bob

and Alice and Nobody compete in the second round, Alice win.